



# Innovation trends in industrial biotechnology: a patent landscape analysis

Pélicissier, P.-M., Grabowska M., Bergamini M.

2025

This document is a publication by the Joint Research Centre (JRC), the European Commission's science and knowledge service. It aims to provide evidence-based scientific support to the European policymaking process. The contents of this publication do not necessarily reflect the position or opinion of the European Commission. Neither the European Commission nor any person acting on behalf of the Commission is responsible for the use that might be made of this publication. For information on the methodology and quality underlying the data used in this publication for which the source is neither Eurostat nor other Commission services, users should contact the referenced source. The designations employed and the presentation of material on the maps do not imply the expression of any opinion whatsoever on the part of the European Union concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries.

#### **EU Science Hub**

<https://joint-research-centre.ec.europa.eu>

JRC139154

EUR 40335

PDF ISBN 978-92-68-28045-4 ISSN 1831-9424 doi:10.2760/8476579 KJ-01-25-306-EN-N

Luxembourg: Publications Office of the European Union, 2025

© European Union, 2025



The reuse policy of the European Commission documents is implemented by the Commission Decision 2011/833/EU of 12 December 2011 on the reuse of Commission documents (OJ L 330, 14.12.2011, p. 39). Unless otherwise noted, the reuse of this document is authorised under the Creative Commons Attribution 4.0 International (CC BY 4.0) licence (<https://creativecommons.org/licenses/by/4.0/>). This means that reuse is allowed provided appropriate credit is given and any changes are indicated.

For any use or reproduction of photos or other material that is not owned by the European Union permission must be sought directly from the copyright holders.

- Cover page illustration, © natrot [1381301983] / stock.adobe.com

How to cite this report: Pélissier, P.-M., Grabowska M., Bergamini M., *Innovation trends in industrial biotechnology: a patent landscape analysis*, Publications Office of the European Union, Luxembourg, 2025, <https://data.europa.eu/doi/10.2760/8476579>, JRC139154.

## Contents

|                                                              |    |
|--------------------------------------------------------------|----|
| Abstract .....                                               | 2  |
| Acknowledgements .....                                       | 3  |
| Key findings.....                                            | 4  |
| 1 Introduction .....                                         | 5  |
| 2 Methodology .....                                          | 6  |
| 2.1 Scope of the study & definitions .....                   | 6  |
| 2.2 Data retrieval.....                                      | 7  |
| 3 Results.....                                               | 9  |
| 3.1 Trends in patenting activities.....                      | 9  |
| 3.2 Patents per technological areas.....                     | 10 |
| 3.3 Revealed Technological Advantage.....                    | 13 |
| 3.4 Source of innovation.....                                | 14 |
| 3.5 Technology transfer and collaborations.....              | 15 |
| 4 Discussion.....                                            | 19 |
| 4.1 Insights from the patent landscape.....                  | 19 |
| 4.1.1 Geographical Trends and Economic Impact .....          | 19 |
| 4.1.2 Technology Specialization and Patent Productivity..... | 20 |
| 4.1.3 Collaboration and Knowledge Transfer.....              | 20 |
| 4.2 Countries-specific particularities .....                 | 20 |
| 4.2.1 South Korea .....                                      | 20 |
| 4.2.2 Singapore .....                                        | 21 |
| 4.2.3 Israel .....                                           | 21 |
| 4.2.4 Switzerland.....                                       | 21 |
| 4.2.5 Belgium .....                                          | 22 |
| 4.3 Follow-up studies .....                                  | 22 |
| 4.4 Limits of the study.....                                 | 23 |
| 5 Conclusion .....                                           | 24 |
| References .....                                             | 25 |
| List of figures.....                                         | 26 |
| Annexes .....                                                | 27 |

## **Abstract**

This report presents a patent landscape analysis investigating the innovation trends within the industrial biotechnology (IB) sector from 2015 to 2020. The study's primary objective is to identify the geographical hotspots of innovation, the key players, and the role of different types of organizations in driving technological advancements in IB. By employing a methodology that includes data retrieval through the Technology Innovation Monitoring (TIM) tool and careful selection of keywords and Cooperative Patent Classification (CPC) terms, the report categorizes patents across five technological areas pertinent to IB. The geographical scope of the analysis encompasses major global players as well as the European Union, providing a broad view of the innovation landscape. The report also introduces an online dashboard to facilitate further analysis and exploration of the data. This study serves as a resource for policymakers, industry stakeholders, and researchers, offering insights that can inform strategic planning and decision-making in the evolving field of industrial biotechnology.

## **Authors**

Pierre-Mathieu PELISSIER

*JRC.B.6 - Industrial strategy, skills and technology transfer*

Marcelina GRABOWSKA, Michela BERGAMINI

*JRC.T.5 - Text and Data Mining*

## **Acknowledgements**

The authors thank Hugo Confraria and Lorenzo Napolitano for their expertise and guidance in the selection of Cooperative Patent Treaty (CPC) terms. Additionally, the authors are thankful to Olivier Eulaerts for his constructive feedback which enhanced the quality of this document.

## Key findings

- The USA remains the top contributor to global patent filings in the Industrial Biotechnology sector, accounting for 49% of total patents from 2015 to 2020, despite a slight decline in its growth rate. China follows with 14%, showing both high growth and a substantial share, highlighting its growing role in global innovation.
- Asian countries, particularly Israel, China, and Singapore, show rapid growth in patent filings, with a compound annual growth rate (CAGRs) above 27%, underscoring Asia's increasing focus on industrial biotechnology and challenging traditional leaders like the USA and Japan.
- European Union patenting data reveals a significant disparity among member states. While several countries demonstrate strong innovation and growth in industrial biotechnology, the overall EU performance is moderate compared to the USA and leading Asian nations. This is largely because the strong performance of leading innovators within the EU is counterbalanced by the slower growth or decline observed in other member economies. This disparity raises important policy considerations regarding whether to prioritize support for existing centers of excellence or to implement broader strategies aimed at levelling up innovation capacity across all member states.
- When normalized by gross domestic product (Purchasing Power Parity) (PPP-GDP), smaller countries like South Korea, Singapore, and Belgium show high patent intensities, indicating efficient innovation systems but the overall EU patent intensity remain low compared to other global leaders.
- The USA leads in all technological sectors analysed in terms of total number of patents. Specific countries excel in certain fields when normalizing the number of patents on PPP-GDP: for instance Sweden in Bioprocess Engineering, Germany in Molecular Diagnostics and Robotics, South Korea in Computational Biology, the UK in Molecular Diagnostics.
- The USA maintains a dominant position across all five technology fields, with Revealed Technology Advantage (RTA) values above 1, indicating a high level of specialization. While China shows rapid growth in patent filings, its RTA suggests it does not specialize strongly in industrial biotechnology. Individual countries like Belgium, Germany, and France display strong specialization in various fields.
- There is a variation in the sources of patents, with universities and research centers leading in some countries (e.g., Singapore, Israel), and private companies in others (e.g., Netherlands, Belgium). This suggests that innovation can thrive in diverse ecosystems, and that the composition of these ecosystems varies from country to country.
- The USA and China are at the forefront of jointly owned patents between public and private institutions. The USA is strongly involved in international collaborations, mainly with the UK and Germany, while EU countries have most international collaborations within the bloc.

# 1 Introduction

Biotechnology and biomanufacturing are increasingly recognized as vital sectors to address societal and environmental challenges, enhance competitiveness, and modernize the economy. In its recent communication related to the “Building the future with Nature” initiative<sup>1</sup>, the European Commission highlights the key role of the Biotechnology and Biomanufacturing sectors in advancing the European Green Deal objectives by providing sustainable solutions for climate change mitigation and adaptation; fostering the sustainable use of natural resources; enhancing food supply and security; and preserving human health. Biotechnology is also identified as a critical technology from an economic security perspective and prioritized in the Strategic Technologies for Europe Platform (STEP)<sup>2</sup> regulation.

The industrial biotechnology (IB) sector, which uses biological systems to produce goods and services, is undergoing a transformation driven by the convergence and maturation of emerging and enabling technologies such as synthetic biology, artificial intelligence (AI), microfluidics, computational biology, and gene editing. These technologies, individually and collectively, have the potential to enable the development of novel products and markets by improving production process efficiency and reducing costs. Understanding the innovation ecosystem surrounding these enabling technologies and their application in IB is therefore critical to informing policy making and strategic planning to strengthen the EU's biotechnology and biomanufacturing competitiveness, and to realize the full potential of these sectors for a greener and more resilient economy. This report aims to address the following goals:

- Mapping the geographical hotspots of innovation in emerging technologies relevant to IB, providing insights into the global distribution of research and development activities.
- Pinpointing key players, including companies, universities, and research institutions, which are driving innovation in IB.
- Exploring the role of distinct types of organizations, such as universities, corporations, and public-private collaborations in fostering innovation in enabling technologies applied to IB.

This report will explore the current state and future trajectory of the IB sector, with a particular focus on the role of emerging technologies. The findings will be of interest to policymakers, industry stakeholders, and researchers looking to understand the dynamics of innovation in this rapidly evolving field.

An online dashboard accompanies this publication to facilitate user analysis and exploration of the underlying data used in the creation of this report.<sup>3</sup>

---

<sup>1</sup> [EC communication on Building the future with nature \(europa.eu\)](https://ec.europa.eu/communication/en/building-the-future-with-nature)

<sup>2</sup> [https://strategic-technologies.europa.eu/index\\_en](https://strategic-technologies.europa.eu/index_en)

<sup>3</sup> [http://www.timanalytics.eu/TimTechPublic/dashboard/index.jsp#/space/s\\_2242?ds=304761](http://www.timanalytics.eu/TimTechPublic/dashboard/index.jsp#/space/s_2242?ds=304761)

## 2 Methodology

### 2.1 Scope of the study & definitions

Industrial biotechnology (white biotechnology): This field involves the application of biological principles and processes to develop sustainable solutions for **industrial** manufacturing and production, including the creation of bio-based products, chemicals, and energy. It uses microorganisms, enzymes, and genetic engineering techniques to optimize industrial processes, minimizing environmental impact and enhancing efficiency. This definition focuses on the **process** (how a product is made) and not on the end product itself.

Emerging (bio)technologies: Emerging biotechnologies and enabling technologies (robotics, AI, microfluidics...) that apply to IB. Examples of such technologies include synthetic biology, engineered biology, gene editing, metabolic engineering, directed evolution, microfluidics applied to the selection of organisms, cell-free biotechnology, nanobiotechnology, cellular agriculture, machine/deep learning, bioinformatics, and data analysis applied to bioprocess and bioengineering.

Geographical scope: The geographical scope of the study encompasses the European Union (EU) and other global leaders in IB, such as Switzerland, South Korea, Japan, China, the UK<sup>4</sup>, the USA, Singapore, and Israel.

Period: 2015–2020. Due to the significant time lag between patent application filing and granting, which varies across patent offices and can take several years, patent data beyond 2020 may be incomplete. Therefore, to ensure data accuracy and completeness, this analysis considers only granted patents up to 2020. The starting point of 2015 was selected based on key technological advancements that have contributed to shaping the IB sector. Notably, the CRISPR-Cas9 gene editing technology, first described in 2012, rapidly gained prominence and demonstrated widespread applicability in 2013 and 2014. This period also marked significant progress in other foundational biotechnologies, such as advancements in genome sequencing towards the 'thousand-dollar human genome', and later, AI-assisted biological predictions gained traction in 2018 with the breakthrough of Google's AlphaFold. By starting in 2015, this analysis captures the synergistic effects of innovations in gene editing, artificial intelligence (notably machine and deep learning), synthetic biology, and other enabling technologies.

Technological areas of interest: We categorized emerging biotechnologies and enabling technologies into major technological domains to highlight countries' specialization. The technological domains were defined to cover as much as possible the technologies of interest while minimizing overlap between them. Their definition is based on previous works refined upon expert consultation<sup>5,6</sup>. The

---

<sup>4</sup> In this analysis, the United Kingdom is always considered individually, even during the period where it was part of the EU (2015–2019). Patents originally assigned to Anguilla, to the Cayman Islands and to the British Virgin Islands are grouped under the "United Kingdom".

<sup>5</sup> Nielsen, J., Tillegreen, C. B., & Petranovic, D. (2020). Innovation trends in industrial biotechnology. *Trends in biotechnology*, 40(10), 1160–1172.

<sup>6</sup> [Europe's Bio Revolution: Innovation examples | McKinsey](#)

resulting technological domains related to industrial biotechnologies and enabling technologies are as follows:

- Synthetic/Engineered biology technologies:
  - These technologies involve designing and building new biological parts, devices, and systems or re-designing existing natural biological systems. They include genome editing tools like CRISPR, TALENs, and ZFNs; synthetic DNA and RNA; metabolic engineering techniques; and the creation of engineered microorganisms and cells with the aim of developing novel bio-based products, optimizing cellular processes, or enhancing the capabilities of biological systems.
- Biological data analysis and computational tools:
  - Technologies associated with the analysis and interpretation of large-scale biological data using computational techniques. This area covers bioinformatics, genomic and sequence data analysis, machine learning, and artificial intelligence applied to biological data. These tools are critical for understanding complex biological systems, predicting outcomes, and designing new biological entities or pathways.
- Robotics, high-throughput assays, and lab automation:
  - Technologies that focus on the use of automation and robotics to accelerate biological research and biotechnological processes. This area includes high-throughput screening, lab-on-a-chip technologies, liquid handling systems, microfluidics, etc. The goal of these technologies is to increase efficiency, precision, and scalability in experiments and processes such as cell sorting, phenotypic screening, and molecular diagnostics.
- Bioprocess engineering:
  - Technologies related to bioprocesses and scaling production: innovations in (bio)reactor design, upstream/downstream processes, and the production of bio-based products.
- Molecular diagnostics and analytical techniques:
  - Technologies for the detection, quantification, and analysis of biological molecules. They include sequencing technologies, bioimaging techniques, and diagnostic assays.

## **2.2 Data retrieval**

### Patent data retrieval:

Patent data was retrieved using the TIM (Technology Innovation Monitoring) tool developed and managed by the JRC - European Commission<sup>7</sup>. We identified for each of the 5 technological

---

<sup>7</sup> [TIM analytics | Knowledge for policy \(europa.eu\)](https://ec.europa.eu/economy_finance/tim/)

domains of interest relevant keywords and Cooperative Patent Classification (CPC) codes. Different approaches were used for their identification; we started from a list of “benchmark” companies recognized as specialized and innovators in the field of industrial biotechnology and related enabling technologies (**Annex 1**). Using natural language algorithms available in the TIM Tool, relevant keywords were extracted from the titles and abstracts of patents retrieved from these companies. We also identified recurrent CPC codes related to these companies’ patent portfolio. We then refined this list via expert consultations and by consulting the existing literature, which also helped us define the sub-sectors covered in this study<sup>8</sup>. The resulting list of keywords and CPC codes associated to each of the technological areas are available in **Annex 2**.

For this study, we retrieved IP5 patent families in the field of IB filed between 2015 and 2020. IP5 patent families are defined as sets of patent applications protecting the same invention, filed in multiple intellectual property (IP) offices. At least one of these filings must be in an office belonging to the IP5 consortium, which includes the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of China (CNIPA), and the United States Patent and Trademark Office (USPTO). Using IP5 patent families ensures higher patent quality and comparability across countries, as these patents represent significant inventions with broader market potential. This approach reduces biases from differing national patenting practices and enhances global innovation insights. For this study, the assessment of national performance in industrial biotechnology is based on a full count of IP5 patent families, assigned to countries according to the applicant's country. This method ensures cross-country comparability by accounting for high-quality patents with international market relevance. Patents were then retrieved using the TIM Analytics tool for each of the 5 technological domains selected by using a dedicated query. For each domain, we selected patents whose title or abstract contain at least one relevant keyword AND one of the CPC codes we identified.

#### GDP data retrieval for normalization:

We considered throughout the study the total patent count per country, and we normalized this data against each country’s Purchasing-Power-Parity-adjusted Gross Domestic Product (PPP-GDP) in current international USD. For all countries except China and Taiwan, this data was retrieved from the World Bank via the UNData website (accessed in July 2024)<sup>9</sup>. For China and Taiwan, this data was retrieved from the International Monetary Fund (IMF)<sup>10</sup>. The value considered for the normalization is the average GDP over the observation period.

---

<sup>8</sup> Carbonell P, Gök A, Shapira P, Faulon JL. Mapping the patent landscape of synthetic biology for fine chemical production pathways. *Microb Biotechnol*. 2016 Sep;9(5):687-95. doi: 10.1111/1751-7915.12401. Epub 2016 Aug 3. PMID: 27489206; PMCID: PMC4993189.

<sup>9</sup> [UNdata | record view | GDP, PPP \(current international \\$\)](#)

<sup>10</sup> [World Economic Outlook \(April 2024\) - GDP, current prices \(imf.org\)](#); last accessed in July 2024

## **3 Results**

### **3.1 Trends in patenting activities**

The European Union faces a competitive global innovation landscape. While the United States remains the dominant force in overall patent granted from 2015-2020, the pace of innovation is now set by others, challenging Europe's strategic standing.

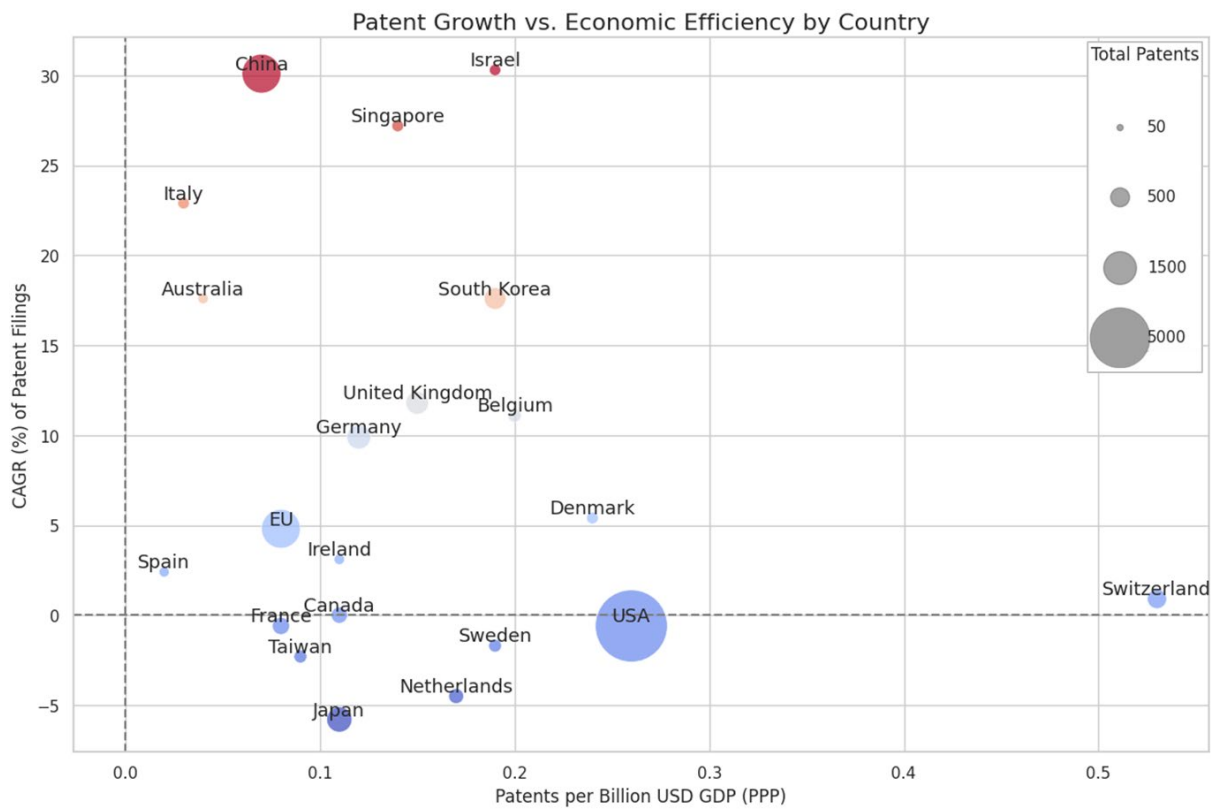
This period is characterized by explosive growth rates outside the EU. Israel leads with a 30.3% Compound Annual Growth Rate (CAGR) in patent filings, followed closely by China (30.1%) and Singapore (27.2%) (Figure 1). Crucially for Europe, China's expansion (its patent output effectively multiplied 3.75 times over these six years) saw it overtake the EU's total patents within this period. This rapid growth, combined with China's already substantial 14.0% global patent share, signals a shift in global innovation leadership. South Korea and Australia also demonstrate robust expansion with CAGRs between 17-23%.

Within Europe, the picture is mixed. While some member states such as Italy, Belgium or Germany show moderate or strong growth, the European Union bloc as a whole records a modest 4.8% CAGR. This signals a slower pace of progress compared to leading global players for the bloc as a whole, and highlights a strong variation in the specialization in the IB field of the bloc's economies. Indeed, major economies like France (-0.6% CAGR) and Sweden saw patent activity decline, mirroring downturns in Japan (-5.8%) and Taiwan.

Despite its own declining growth, the USA still contributed a dominant 49.2% of all global patents. However, China's rapid emergence is reshaping the global patent map. While smaller economies like Israel and Singapore contribute less than 1% to the global total, their rapid growth from a relatively small base showcases powerful national innovation ecosystems.

This highlights a critical challenge for the EU: its moderate growth and internal disparities risk ceding ground to more dynamic innovation hubs. To secure its future competitiveness and strategic autonomy, Europe must confront these trends, foster a more integrated and ambitious innovation ecosystem, and ensure its patent output matches its economic potential.

**Figure 1:** Patent growth compared to economic efficiency per country over the 2015-2020 period for the sector of IB. The complete table used to produce this figure is visible in Annex 3.

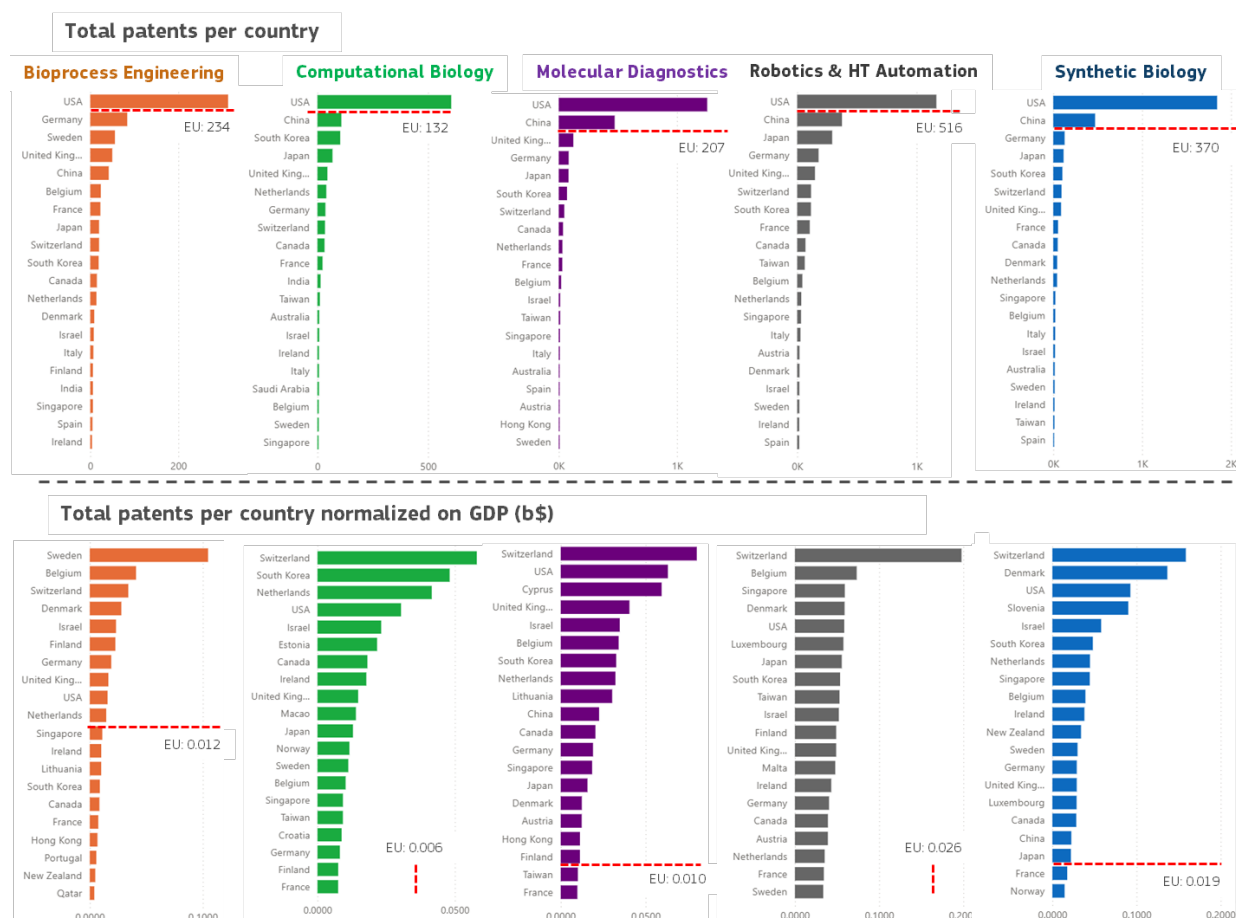


Source: TIM Analytics Tool [TIM analytics | Knowledge for policy](#)

### 3.2 Patents per technological areas

An analysis of patent activity across specific technological areas reveals distinct global and regional leaders (Figure 2). The United States consistently ranks first in total patent counts across all categories, underscoring its dominant innovation ecosystem. The European Union shows strong performance, ranking second globally in Bioprocess Engineering, Computational Biology, and Robotics & High-Throughput Automation, and third in Synthetic Biology and Molecular Diagnostics.

**Figure 2:** Countries rank in terms of total patent count (top) and patent intensity (total counts or per PPP-GDP, bottom) over the observed period and per technological area. Details of the specific patent affiliates (private companies, research centers, universities...) are available per technological field on the dashboard provided with this publication<sup>11</sup>.



Source: TIM Analytics Tool [TIM analytics | Knowledge for policy](http://www.timanalytics.eu/TimTechPublic/dashboard/index.jsp#/space/s_2242?ds=304761)

## Bioprocess Engineering

In Bioprocess Engineering, Sweden emerges as a global leader in patent intensity, boasting more than twice the patents per unit of GDP compared to second-ranked Belgium. This efficiency is driven by companies such as GE Healthcare Biosciences, Cytiva, and Sartorius, complemented by robust government initiatives like the Testa Center. Within Europe, Germany distinguishes itself through its substantial total patent count, propelled by industry giants including Sartorius, Merck, Boehringer, and Eppendorf.

Despite these individual successes, the European Union as a collective performs moderately low on the global stage in bioprocess engineering. This paradox is explained by a disparity among member states: while countries like Sweden, Belgium, Denmark, Finland, Germany, and the Netherlands

<sup>11</sup> [http://www.timanalytics.eu/TimTechPublic/dashboard/index.jsp#/space/s\\_2242?ds=304761](http://www.timanalytics.eu/TimTechPublic/dashboard/index.jsp#/space/s_2242?ds=304761)

stand out as strong innovation hubs and top global ranks, the overall EU average is significantly diluted by the comparatively lower patenting activity from other major economies, such as France, Italy, or Poland. This structural imbalance highlights that the EU's innovation landscape in this domain is characterized by pockets of excellence coexisting with regions that are lagging. Addressing this calls for a dual policy approach: either strategically reinforcing the leading innovation ecosystems to maximize their global impact, or implementing targeted investment and capacity-building initiatives in underperforming member states to foster a more balanced and robust Union-wide innovation front.

### **Computational Biology**

South Korea leads globally in both patent counts and intensity in computational biology due notably to the activity of the Korea Advanced Institute for Science and Technology (KAIST) and companies like GC Genome Corp, Oncocross, HEM Pharma, and SeeGene. In Europe, the Netherlands and Germany take leading roles. The Netherlands' contributions are driven by companies like Koninklijke Philips N.V., while Germany's patents stem from entities such as BASF SE, Curetis GmbH, KWS SAAT, Roche Diagnostics, and Siemens Healthcare.

### **Molecular Diagnostics**

Switzerland leads in patents per unit of GDP in molecular diagnostics, outperforming the USA. Germany, however, leads in Europe for total patent counts, supported by companies like Roche Diagnostics, Qiagen GmbH, Bosch GmbH, Siemens Healthcare, and Miltenyi Biotech. Globally, the UK ranks third in total patents, thanks to its strong gene sequencing ecosystem. Key contributors include Illumina Cambridge and Oxford Nanopore Technologies, both active in DNA sequencing technology. The UK's long history of innovation in this field, from the discovery of DNA's structure at the University of Cambridge to Sanger sequencing, has fostered a robust genomics sector. Cyprus also ranks highly in patent intensity, largely due to NIPD Genetics, a leader in genetic testing solutions.

### **Robotics & High-Throughput Automation**

Switzerland ranks first globally in patent intensity in robotics and high-throughput automation, led by companies such as Tecan Trading, Hamilton Bonaduz, and Hoffman-La Roche. Germany leads in Europe for total patent counts, with contributions from companies like Bosch GmbH, Eppendorf, Brand GmbH, and the Fraunhofer Institute. These organizations are notably active in advancing automated liquid handling and lab automation technologies. Similarly as for other sectors, while Europe as a whole is performing low on the global stage, a few countries top the ranks such as Belgium, Denmark, Luxemburg or Finland indicating a strong innovation disparity within the bloc.

### **Synthetic Biology**

In synthetic biology, Switzerland leads globally in patent intensity, followed by Denmark and the USA. Germany and France are Europe's top contributors in total patent counts. Germany's biggest innovators in this field in our study are KWS SAAT, BASF SE, the Fraunhofer and Max Planck Institutes, and BioNTech. France's efforts are driven by research organizations such as INSERM and CNRS, along with private companies like Collectis and Total Energies.

### 3.3 Revealed Technological Advantage

We measured the Revealed Technological Advantage (RTA) of the EU and its largest global competitors in the field of industrial biotechnology (Figure 3). The RTA indicates the extent to which a country specializes in a particular technology sector relative to other countries, hereby “revealing” where a country has a stronger technological capability based on its patent activity in a specific area compared to global averages. We calculate the RTA index by comparing a country’s share of IP5 patents in a specific technology to its overall share of IP5 patents across all technologies. This ratio is then normalized by the global share of IP5 patents in the same technology, ensuring that a country’s specialization is assessed in a global context rather than being limited to a specific set of economies. An RTA value greater than 1 indicates that a country has a higher relative concentration of patents in a given technology compared to the global average, signifying a revealed technological advantage.

**Figure 3:** Revealed technological advantage for the EU and global competitors (AU: Australia; CA: Canada ; CH: Switzerland ; CN: China ; EU: European Union (27 members) ; UK: United Kingdom ; IL: Israel ; JP: Japan ; KR: South Korea ; SG: Singapore ; TW: Taiwan ; US: United States of America).

|                                            | AU   | CA   | CH   | CN   | EU   | UK   | IL   | JP   | KR   | SG   | TW   | US   |
|--------------------------------------------|------|------|------|------|------|------|------|------|------|------|------|------|
| Biological Data Analysis and Computational | 0.46 | 1.19 | 0.43 | 0.21 | 0.47 | 1.11 | 0.47 | 0.20 | 0.44 | 3.15 | 0.17 | 1.07 |
| Bioprocess Engineering                     | 0.37 | 0.28 | 1.24 | 0.04 | 1.02 | 1.96 | 0.51 | 0.10 | 0.21 | 0.91 | 0.05 | 1.08 |
| Engineered Biology<br>Synthetic Biology    | 0.68 | 0.82 | 0.72 | 0.14 | 0.39 | 0.48 | 0.64 | 0.12 | 0.38 | 1.37 | 0.07 | 1.40 |
| Molecular Diagnostics                      | 0.23 | 0.65 | 0.17 | 0.51 | 0.41 | 1.97 | 0.64 | 0.13 | 0.21 | 1.70 | 0.12 | 1.28 |
| Robotics & High-Throughput Automation      | 0.09 | 0.54 | 1.37 | 0.39 | 0.59 | 0.68 | 0.56 | 0.42 | 0.35 | 1.86 | 0.32 | 1.04 |

Source: TIM Analytics Tool [TIM analytics](#) | Knowledge for policy

When analysing technological specialization, the USA confirms<sup>12</sup> its dominant position, with RTA index values exceeding one across all five technology fields. Singapore, the UK, and, to a lesser degree, Switzerland also stand out as global hotspots in most of the technology areas of interest, with RTA index values indicating strong specialization paired with remarkable growth in patent filings over the study period. Intriguingly, despite China’s significant growth rate in patent filings, its RTA index values do not reflect a corresponding level of specialization in these five technology areas, implying that it under-specializes in these fields relative to its broader technological efforts. This may suggest that China allocates its research and development resources more evenly across different fields or it has a stronger focus on other technology areas. Singapore, Switzerland, and the UK exhibit high RTA values across multiple domains, positioning them as global hotspots for niche specialization. Switzerland’s RTA in Molecular Diagnostics and Singapore’s in Computational Biology stand out as strong cases of targeted innovation.

<sup>12</sup> Van Looy, B., Grabowska, M., Vlaisavljevic, V., Lecocq, C., & Debackere, K. (2024). Growth of biotech clusters over several decades through pioneering, variety and entrepreneurial science. *Nature Biotechnology*, 42(1), 20-25.

The EU collectively shows specialization only in Bioprocess Engineering, with an RTA index above one in this domain (Figure 3). In the remaining four fields, the EU underperforms compared to the global average, despite strong contributions from individual member states (Figure 4). Among the EU Member States with at least 10 patents in the IB subdomain, Belgium and Portugal exhibit the highest RTA values, particularly in Bioprocess Engineering (15.41 and 13.94, respectively) and Robotics & High-Throughput Automation (15.06 for Portugal). These high values indicate a strong specialization in these fields. Denmark (6.87 in Engineered/Synthetic Biology) and Spain (5.03 in Robotics & High-Throughput Automation) also show notable specialization in specific areas. Meanwhile, Germany, France, Italy, and the Netherlands display RTA values mostly around or slightly above 1 across multiple fields, suggesting a more balanced distribution of technological advantage.

**Figure 4:** Revealed technological advantage for EU member states. Only countries with at least 10 patents in each industrial biotechnology sub-technology are presented in this figure (AT: Austria ; BE: Belgium ; DE: Germany ; DK: Denmark ; ES: Spain ; FI: Finland ; FR: France ; IE: Ireland ; IT: Italy; NL: The Netherlands ; PT: Portugal ; SE: Sweden).

|                                         | AT   | BE    | DE   | DK   | ES   | FI   | FR   | IE   | IT   | NL   | PT    | SE    |
|-----------------------------------------|------|-------|------|------|------|------|------|------|------|------|-------|-------|
| Biological data analysis and computatio | -    | -     | 0.83 | -    | -    | -    | 1.28 | -    | -    | 4.75 | -     | -     |
| Bioprocess engineering                  | -    | 15.41 | 3.50 | -    | -    | -    | 2.35 | -    | 2.47 | 3.61 | -     | 13.94 |
| Engineered/Synthetic biology            | -    | 3.70  | 1.27 | 6.87 | 2.55 | -    | 1.25 | 3.13 | 1.27 | 1.87 | -     | 0.80  |
| Molecular diagnostics                   | -    | 4.17  | 1.04 | -    | 2.61 | -    | 1.00 | -    | 0.84 | 2.16 | -     | 0.92  |
| Robotics & High-Throughput Automation   | 2.92 | 7.15  | 2.01 | 3.53 | 5.03 | 1.82 | 2.97 | 4.78 | 1.87 | 2.24 | 15.06 | 1.61  |

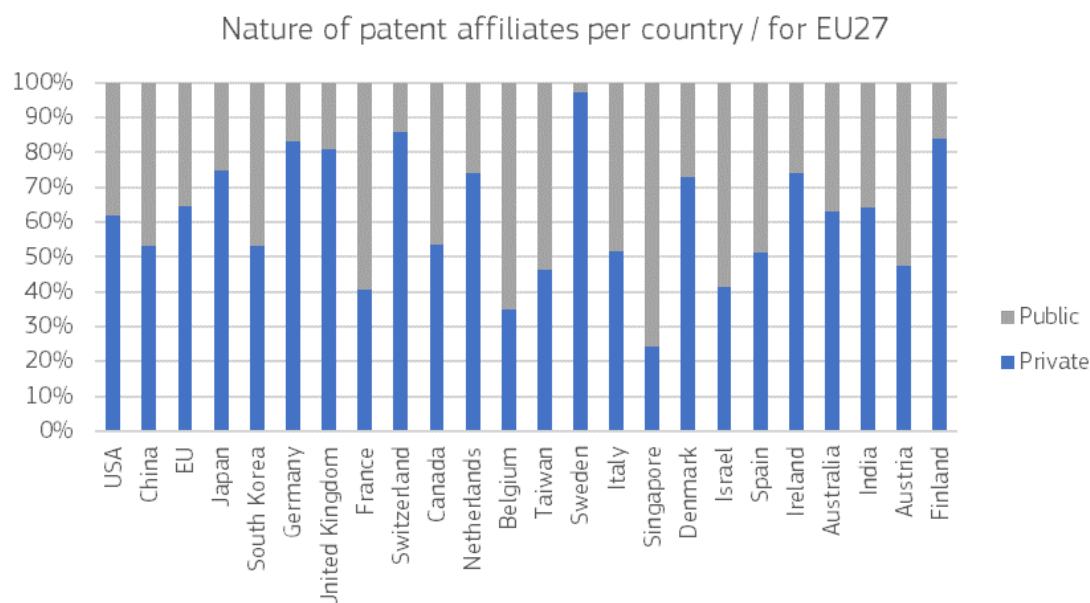
Source: TIM Analytics Tool [TIM analytics | Knowledge for policy](#)

### 3.4 Source of innovation

An analysis of patent affiliation reveals a broadly similar distribution of patenting organizations in the EU, USA, and China, with approximately 60% of patents attributed to private companies and 40% to public sector institutions (e.g., hospitals, research centres, and universities).<sup>13</sup> However, Sweden stands out as an exception, with most patents originating from private companies. This unique distribution is likely influenced by Sweden's "Professor's Privilege," which grants university researchers full ownership of their intellectual property. As patents owned by individuals were excluded from this analysis, there may be a data gap for Sweden (Figure 5).

<sup>13</sup> The identification of the type of actor is based on the name and address information on patents and publication, following the sector allocation method described [in Eurostat, Patent Statistics at Eurostat: Methods for Regionalisation, Sector Allocation and Name Harmonisation \(European Union, 2011\)](#).

**Figure 5:** Distribution of patent affiliations by type (company, public sector) for the 25 most patent-productive countries and the EU, presented as a percentage over the observed period.



Source: TIM Analytics Tool [TIM analytics](#) | Knowledge for policy

Significant disparities exist in the balance of patenting activity between the public and private sectors across countries. In France and Belgium, innovation is predominantly driven by public sector institutions, which account for 60% of patents, with private companies contributing the remaining 40%. Conversely, countries like the UK, Germany, and Switzerland demonstrate a stronger reliance on private-sector innovation, with over 80% of patents originating from companies.

Singapore, despite being a leading innovator, exhibits a unique profile where only 25% of patents have companies as assignees, while 75% are linked to publicly funded organizations. This contrasts sharply with countries such as Denmark, the Netherlands, Germany, the UK, the USA, and Switzerland, where private companies dominate patent filings.

These differences reflect heterogeneity in the structure of national innovation ecosystems. Countries like Singapore, Belgium, and France rely more heavily on public-sector contributions to innovation in this specific domain, whereas others, such as Germany, Switzerland, and the USA, are more dependent on private-sector activity. This suggests that both public and private investments play roles in driving innovation in industrial biotechnology (IB), although the balance varies by country.

### 3.5 Technology transfer and collaborations

Analysing public-private co-owned patents, those listing at least one applicant from the public sector (e.g., universities, research centres, hospitals) alongside at least one private-sector applicant (e.g., companies), provides insights into trends in knowledge and technology transfer (Figure 6).

**Figure 6:** private-public patents co-affiliations.

| Country        | % of patents with at least one private-public co-affiliation | % of patents with at least two co-affiliates | Total patents |
|----------------|--------------------------------------------------------------|----------------------------------------------|---------------|
| Spain          | 45.0                                                         | 32.5                                         | 40            |
| France         | 34.9                                                         | 40.0                                         | 235           |
| Italy          | 29.2                                                         | 29.2                                         | 72            |
| South Korea    | 22.2                                                         | 15.6                                         | 410           |
| Japan          | 20.2                                                         | 16.0                                         | 580           |
| China          | 14.6                                                         | 16.3                                         | 1469          |
| EU             | 10.5                                                         | 5.5                                          | 1459          |
| Canada         | 6.9                                                          | 17.3                                         | 202           |
| India          | 5.4                                                          | 5.4                                          | 37            |
| USA            | 5.2                                                          | 12.1                                         | 5184          |
| Germany        | 5.1                                                          | 16.8                                         | 511           |
| Singapore      | 3.9                                                          | 13.0                                         | 77            |
| Israel         | 3.0                                                          | 13.6                                         | 66            |
| Denmark        | 2.6                                                          | 13.2                                         | 76            |
| Taiwan         | 1.9                                                          | 13.6                                         | 103           |
| United Kingdom | 1.8                                                          | 19.1                                         | 455           |
| Belgium        | 1.7                                                          | 19.8                                         | 116           |
| Netherlands    | 1.2                                                          | 9.3                                          | 162           |
| Switzerland    | 0.0                                                          | 22.3                                         | 310           |
| Sweden         | 0.0                                                          | 20.8                                         | 101           |
| Australia      | 0.0                                                          | 13.3                                         | 45            |
| Ireland        | 0.0                                                          | 4.4                                          | 45            |
| Austria        | 0.0                                                          | 12.9                                         | 31            |
| Finland        | 0.0                                                          | 3.8                                          | 26            |

Source: TIM Analytics Tool [TIM analytics | Knowledge for policy](#)

### Co-ownership and public-private partnerships

An examination of co-owned patent data provides insights into the collaborative dynamics of national innovation systems. This analysis highlights significant variations in partnership structures.

Spain exhibits the highest proportion of public-private co-owned patents (45%), signalling a robust emphasis on academia-industry collaboration. France (34.9%) and Italy (29.2%) also demonstrate substantial engagement in this model. Conversely, Switzerland, Sweden, Finland, and Austria register no instances of public-private patent co-ownership within the covered timeframe. This suggests that innovation in these economies predominantly occurs within distinct sectorial boundaries.

Beyond the public-private dimension, a broader analysis of multi-entity patent co-ownership reveals extensive collaborative networks. France (40%) and Italy (29.2%) maintain high rates, indicative of well-established collaborative infrastructures that may encompass diverse entity types or multiple contributors within a single sector. Smaller economies such as Belgium (19.8%) and the United Kingdom (19.1%) also exhibit elevated rates of multi-ownership collaboration, attesting to dynamic innovation ecosystems despite comparatively lower absolute patent counts.

Surprisingly, Switzerland (22.3%) and Sweden (20.8%), despite their lack of public-private co-ownership, demonstrate significant rates of multi-ownership collaboration. This suggests that their

collaborations are primarily intra-sectoral, involving partnerships such as private-private ventures within industry or public-public collaborations within academia, rather than cross-sectoral interactions.

Despite their substantial overall patent output, the USA (5.2%) and China (14.6%) register comparatively low proportions of public-private co-owned patents. Their multi-ownership collaboration rates (12.1% and 16.3%, respectively) are modest in comparison to leading European nations. This may signify a reliance on well-resourced, autonomous research and development within both public and private sectors with a lower emphasis on cross-sectoral partnerships.

### **International collaborations**

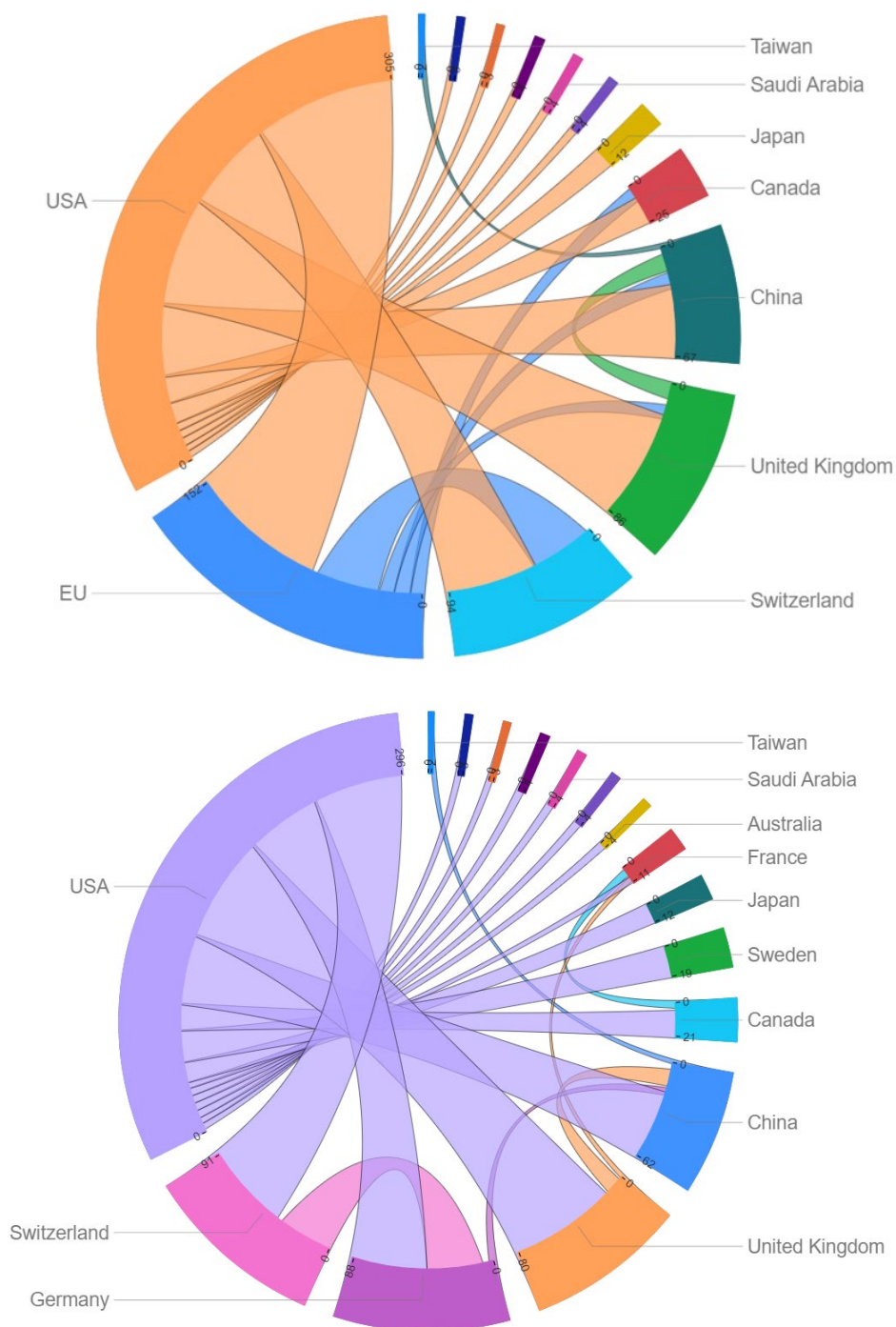
International collaborations represent a critical dimension of patent co-ownership, extending beyond domestic innovation boundaries. Figure 7 illustrates the geographical distribution of patents involving co-ownership, highlighting key global partnerships.

The USA consistently ranks as the preeminent nation in terms of international patent co-ownership during the observed period. Its primary international collaborators include the EU, followed by Switzerland, the UK, China, Canada, and Japan, in descending order of relevance. The EU, conversely, demonstrates a significant reliance on the USA, with over 50% of its international co-owned patents involving American entities, and approximately 25% with Switzerland, with smaller proportions attributed to the UK, China, and Canada.

An examination of individual EU Member States reveals that most USA-EU co-ownerships are concentrated in Germany (48 of 86 jointly held patents) and Sweden (20 patents). Furthermore, Germany functions as a pivotal intermediary in EU-Switzerland (36 of 39 patents) and EU-China (5 of 10 patents) collaborations.

Intra-EU co-ownership patterns frequently reflect strong geographical and regional affiliations. For instance, Belgium's international co-owned patents demonstrate a 50% share with France (2) and The Netherlands (2), respectively. Similarly, Finland's sole international patent is co-owned with Norway. Despite the presence of these regional linkages, the EU27 countries predominantly engage in patent co-ownership with external partners, including the USA, Switzerland, and China. This underscores the EU's outward-looking approach to international collaboration, with Germany notably serving as a key nexus between European innovation systems and major global players.

**Figure 7:** Representation of international co-affiliations with EU27 countries grouped (top) or individually (bottom). These graphs represent the number of times countries are co-owners on a patent with each other's.



Source: TIM Analytics Tool [TIM analytics | Knowledge for policy](#)

## 4 Discussion

### 4.1 Insights from the patent landscape

#### 4.1.1 Geographical trends and economic impact

The United States' dominance in the IB patent landscape is consistent with its long-standing leadership in biotechnological innovation and substantial investment in research and development. The surge in patent filings from China, surpassing Japan and the EU, signals that its long-term investments in technological capabilities and commitment to becoming a leader in high-tech industries is starting to deliver. South Korea's ascension is indicative of its strategic investment in emerging technologies, particularly in the fields of computational biology and high-throughput automation.

It is noteworthy that when patent counts are normalized by PPP-GDP, smaller countries like South Korea, Singapore, and Belgium exhibit high patent intensities, suggesting efficient innovation systems that generate a significant number of patents relative to their economic size. This shows the importance of targeted investment strategies in supporting an environment conducive to innovation, regardless of a country's size. The EU's lower patent intensity compared to other global leaders points to potential areas for policy intervention to stimulate innovation.

The varied distribution of patent affiliations across countries reflects different national policies and cultural attitudes towards innovation. In countries where public institutions hold a significant share of patents, such as France, Singapore and Belgium (Figure 5), there may be more robust support for fundamental research and an emphasis on knowledge generation within the academic sector. In countries with such patenting dynamics, it underlines the importance of developing or maintaining a well-functioning technology-transfer capability to transfer academic-based research into innovation. The European Union is known to generate strong research and innovation, but weaker in transferring it to the industry, often for the benefit of other countries<sup>14</sup>. In contrast, the high proportion of private company-affiliated patents in countries like Denmark and the Netherlands suggests a business environment that is conducive to commercial innovation and a strong presence of mature biotechnology industries, and their dynamism within the European Union may serve as a case study for other countries. The "Professor Privilege" in Sweden exemplifies how legal frameworks can influence the source of innovation, encouraging direct commercialization by researchers.

---

<sup>14</sup> Nagar, Jay Prakash, Stefano Breschi, and Andrea Fosfuri. "ERC science and invention: Does ERC break free from the EU Paradox?." *Research Policy* 53.8 (2024): 105038.

### **4.1.2 Technology specialization and patent productivity**

Despite the European Union's moderate overall performance in key technological areas like bioprocess engineering, the success of individual member states highlights a strategic path forward. For instance, Sweden's leadership in bioprocess engineering demonstrates the power of national specialization, fuelled by historical strengths, targeted funding, and key industry players such as GE Healthcare Biosciences and the Testa Center.

This pattern of specialized excellence within certain EU countries, contrasting with the Union's overall lower global standing due to internal disparities, suggests a strategic direction. Rather than striving for universal excellence across all technologies in every member state, an effective EU policy might involve identifying and nurturing distinct technological strengths within individual member states. This approach, coupled with robust promotion of cross-border cooperation, can build complementary innovation ecosystems across the Union, allowing the EU to leverage its diverse expertise to collectively enhance its global competitiveness in these critical sectors.

### **4.1.3 Collaboration and knowledge transfer**

The success of the USA, France, and China in fostering public-private partnerships, as reflected in their significant number of co-owned patents (Figure 6 and 7), highlights the critical role of cross-sector collaboration in driving innovation. The USA, in particular, has successfully positioned itself as a global leader in international co-affiliations, with key partnerships with the EU, Switzerland, and China. Within the EU, Germany and Sweden account for the majority of co-affiliated patents with the USA, demonstrating their leadership in facilitating global research partnerships.

In contrast, countries like Switzerland and Sweden exhibit high levels of internal collaboration (patents with multiple co-affiliates within either the public or private sector) but limited public-private partnerships. This suggests that while these countries have robust innovation ecosystems, opportunities for bridging academic research with industry may remain underexplored. For the EU overall, the low percentage of public-private patents underscores the need for stronger mechanisms to facilitate collaboration and transfer knowledge into industrial applications.

## **4.2 Countries-specific particularities**

We found a few global leaders that stood out in our analysis, particularly in regards with their innovation landscape related to the size of their economy or with their dynamic growth over the observed period. The dynamism and growth of these specific countries in the field of industrial biotechnologies warrant advanced case-studies to highlight past and current policy measures dedicated to this industrial sector. We propose below a brief overview of the policy-related reasons that might have influenced these countries' strength in this area, but further, comprehensive, investigations beyond this patent landscape analysis are required to fully understand how specific policies have fostered a dynamic innovation environment in these countries.

### **4.2.1 South Korea**

South Korea emerged as a global leader in many aspects of IB, and in particular in computational biology (Figure 2 and 3), as highlighted in our patent landscape analysis. South Korea is a global leader in innovation according to the Innovation Scoreboard 2024 published by the European

Commission, ranking first among the global competitors of the EU overall<sup>15</sup>. This success is attributed to both systemic factors and targeted biotech policies. Since 2006, South Korea's "Bio-Vision 2016" plan has set specific biotech and bioeconomy goals, with a later focus on bioindustries and pharmaceuticals<sup>16</sup>. The 2012 "Strategy for promotion of industrial biotechnology" and the 2013 "3rd Basic Plan for Science and Technology" further prioritized biotech innovation, fostering technology transfer, cross-sector alliances, and biobased standards. The Ministry of Science and ICT's Bio-Venture Support Program backs life sciences startups<sup>17</sup>, complemented by R&D tax credits, regulatory streamlining, and IP protection to attract investment. In 2023, the Green Bio Act was introduced to boost biotech in agriculture through development plans, research institutes, startup support, and "green bio" data promotion<sup>18</sup>. Finally, the "National Synthetic Biology Initiative" focuses on innovation systems, bio-manufacturing, startup support, and international collaboration, aiming to develop technology roadmaps and train professionals, underpinned by a supporting legal framework<sup>19</sup>.

### 4.2.2 Singapore

Singapore's focus on alternative proteins and cultivated meat reflects its strategic response to local food security challenges, as observed in its high patent intensity in industrial biotechnology (Figure 2). Its strong reliance on public institutions and government-backed initiatives (Figure 5) emphasizes the critical role of centralized planning in its innovation ecosystem. However, its relatively low rate of international co-affiliations suggests that while Singapore excels in domestic innovation, opportunities for broader global partnerships may exist to further enhance its impact.

### 4.2.3 Israel

Israel's biotech sector has flourished due to strategic investments in education, government support, and industry partnerships. By allocating around 6% of its GDP to education<sup>20</sup>, Israel has created a skilled workforce that attracts global corporations like Pfizer and General Electric Healthcare to establish R&D centers. The Ministry of Innovation, Science and Technology prioritizes sectors such as bio-convergence and foodtech, while entities like the Israel Biotech fund and Start-Up Nation Central provide financial support and facilitate networking<sup>21</sup>.

#### 4.2.4 Switzerland

Switzerland is a renowned hub for innovation in industrial biotechnology, supported by stable politics, a strong IP system, world-class universities, and excellent quality of life, all of which attract

<sup>15</sup> [European innovation scoreboard - European Commission \(europa.eu\)](https://ec.europa.eu/innovation/scoreboard/)

<sup>16</sup> Hyeon BH, Kim HY, Lee CM, Moon SH, Kim ES, Lee SY. Bio-Vision 2016: the second national framework plan for biotechnology promotion in Korea. *Biotechnol J*. 2008 May;3(5):591-600. doi: 10.1002/biot.200700248. PMID: 18446865.

<sup>17</sup> ICT ministry holds meeting on bio venture firms | Yonhap News Agency (yna.co.kr)

<sup>18</sup> South Korea: Act to Promote Green Bio Industry Published | Library of Congress (loc.gov)

<sup>19</sup> Press Releases - 과학기술정보통신부 > ([msit.go.kr](http://msit.go.kr))

<sup>20</sup> [Israel - Public Spending On Education, Total \(% Of GDP\) - 2024 Data 2025 Forecast 1973-2020 Historical \(tradingeconomics.com\)](https://tradingeconomics.com)

<sup>21</sup> Ministry of Innovation determines Israel's national R&D priorities Ministry of Foreign Affairs ([www.gov.il](http://www.gov.il))

global talents<sup>22</sup>. Key industry players like Novartis, Hoffmann-La-Roche, CRISPR-Therapeutics, and Lonza, contribute to its leadership, while Swissmedic streamlines drug approvals more efficiently than other agencies worldwide<sup>23</sup>, enhancing the biopharmaceutical sector's competitiveness and interest of private companies to locate their activities in the country. The Swiss government's life sciences strategy includes corporate tax rates and incentives for manufacturing and R&D, introduced in new tax legislation in 2020, along with specific taxation for IP income.

#### **4.2.5 Belgium**

Belgium possesses an innovation ecosystem dedicated to industrial biotechnology, the Flanders Biobased Valley, which hosts the Bio Base Europe Pilot Plant in Ghent. Initiatives like Catalisti and GreenWin further promote sustainability in the chemical and biotech sectors.

Government support through tax incentives, grants, and funding programs bolsters Belgium's high-tech R&D, encouraging domestic and international investment. The patent income deduction specifically incentivizes research commercialization. Belgium's strategic location in Europe provides additional advantage, offering biotech companies easy access to markets and bio-based feedstocks, which is crucial for the industry's growth.

### **4.3 Follow-up studies**

Based on our report's findings, several follow-up studies can build upon the key findings and address the gaps identified by analysing specific areas more in-depth, assess policy impacts, and explore strategic opportunities for innovation growth.

- A detailed analysis could evaluate the effectiveness of specific policies implemented in countries like South Korea, Singapore, and Belgium that have led to high patent intensities.
- In-depth case studies of countries or regions with high patent productivity, particularly those with efficient innovation systems relative to their economic size, could identify best practices, successful models for university-industry collaboration. The role of legal frameworks like the "Professor Privilege" in Sweden or other incentives such as patent income deductions and tax benefits on innovation activities could be a further area of in-depth enquiry.
- Understanding the factors influencing the distribution of patents among public and private sectors within different countries could provide more insights into how the balance of innovation sources impacts overall productivity.
- The online dashboard is dynamic and can be updated easily, reflecting on the most recent trends in patenting activity. It may also be modified to capture technological domains or technologies not covered in this analysis.

---

<sup>22</sup> [What's the secret to Switzerland's status as a global talent hub? | World Economic Forum \(weforum.org\)](https://www.weforum.org/articles/2019/01/24/What-s-the-secret-to-Switzerland-s-status-as-a-global-talent-hub/?_hpid=article-topics-button-switzerland&_hpt=article-topics-button-switzerland)

<sup>23</sup> Rodier, C., et al. "New drug approvals in six major authorities 2010–2019: focus on facilitated regulatory pathways and internationalisation." *R&D Briefing* 77 (2020): 1–24.

## 4.4 Limits of the study

- This report is descriptive and serves as foundation for future studies.
- Relying solely on patent data to gauge innovation may not provide a complete picture. Not all innovations are patented, and patent counts do not necessarily reflect the commercial viability or impact of the technologies. Additionally, the complexity of patent classifications may lead to under- or over-representation of certain technologies. Moreover, organizations have different strategies for protecting their intellectual property, with some favouring trade secrets or rapid market entry over patent filings. This study does not account for such strategic variations, which can influence the number and type of patents filed.
- The analysis is confined to patent data between 2015 and 2020. Trends or innovations emerging after this period have not been captured, omitting recent developments that could influence the current state of the sector.
- The study focuses on emerging technological fields within industrial biotechnology, such as synthetic biology and AI. Other relevant technologies that did not fall within the defined scope may also play a critical role in the sector's transformation but were not analysed here.
- The technologies are found using a set of keywords and patent CPC terms. The addition or absence of certain terms can influence the outcome.
- There is a time lag between technological innovation and patent filing, as well as between patent filing and publication, meaning that the study may not fully reflect the most innovative advancements at the time of analysis.
- While the study examines co-owned patents, the depth of these collaborations and their impact on innovation could not be fully assessed as the nature of the relationships between co-owners and the quality of their interactions are not reflected in patent data alone.
- While the study highlights the role of public-private collaborations, it does not analyse the effectiveness of these collaborations in driving innovation or their long-term sustainability.

## 5 Conclusion

This patent landscape analysis examined innovation trends within the industrial biotechnology (IB) sector from 2015 to 2020, identifying geographical hotspots, key players, and organizational roles. Our findings confirm the United States and Switzerland as dominant forces in IB and highlights the rapid ascent of Asian countries, particularly China, which has shown remarkable growth in patent filings, surpassing the EU and Japan.

While the European Union demonstrates overall growth, its pace is more moderate compared to the US and key Asian nations. This can be attributed, in part, to the fact that while several smaller European countries exhibit strong patent intensity and efficient innovation systems, the EU's overall performance is influenced by larger economies that may not specialize as heavily in specific IB areas. Nevertheless, the analysis of Revealed Technological Advantage (RTA) reveals the USA's broad technological advantage across IB, while countries like Switzerland and the UK show specialization in specific high-value areas like Molecular Diagnostics and Synthetic Biology. The EU demonstrates specialization primarily in Bioprocess Engineering.

In conclusion, this patent landscape analysis of industrial biotechnology between 2015 and 2020 underscores the dynamic nature of the field. While the USA maintains a leading position, the rapid growth in Asia coupled with the specialized strengths of various European nations highlights a shifting global innovation landscape. Understanding these geographical trends, technological specializations, and collaboration patterns is crucial for policymakers and industry stakeholders aiming to support innovation and ensure continued advancements in this sector. Future studies could explore deeper into specific technological sub-sectors and the impact of evolving regulatory frameworks.

## References

1. [EC communication on Building the future with nature \(europa.eu\)](#)
2. [https://strategic-technologies.europa.eu/index\\_en](https://strategic-technologies.europa.eu/index_en)
3. [http://www.timanalytics.eu/TimTechPublic/dashboard/index.jsp#/space/s\\_2242?ds=304761](http://www.timanalytics.eu/TimTechPublic/dashboard/index.jsp#/space/s_2242?ds=304761)
4. In this analysis, the United Kingdom is always considered individually, even during the period where it was part of the EU (2015-2019). Patents originally assigned to Anguilla, to the Cayman Islands and to the British Virgin Islands are grouped under the “United Kingdom”.
5. Nielsen, J., Tillegreen, C. B., & Petranovic, D. (2020). Innovation trends in industrial biotechnology. *Trends in biotechnology*, 40(10), 1160-1172.
6. [Europe’s Bio Revolution: Innovation examples | McKinsey](#)
7. [TIM analytics | Knowledge for policy \(europa.eu\)](#)
8. Carbonell P, Gök A, Shapira P, Faulon JL. Mapping the patent landscape of synthetic biology for fine chemical production pathways. *Microb Biotechnol*. 2016 Sep;9(5):687-95. doi: 10.1111/1751-7915.12401. Epub 2016 Aug 3. PMID: 27489206; PMCID: PMC4993189.
9. [UNdata | record view | GDP, PPP \(current international \\$\)](#)
10. [World Economic Outlook \(April 2024\) - GDP, current prices \(imf.org\)](#); last accessed in July 2024
11. [http://www.timanalytics.eu/TimTechPublic/dashboard/index.jsp#/space/s\\_2242?ds=304761](http://www.timanalytics.eu/TimTechPublic/dashboard/index.jsp#/space/s_2242?ds=304761)
12. Van Looy, B., Grabowska, M., Vlasisavljevic, V., Lecocq, C., & Debackere, K. (2024). Growth of biotech clusters over several decades through pioneering, variety and entrepreneurial science. *Nature Biotechnology*, 42(1), 20-25.
13. The identification of the type of actor is based on the name and address information on patents and publication, following the sector allocation method described in [Eurostat. Patent Statistics at Eurostat: Methods for Regionalisation, Sector Allocation and Name Harmonisation \(European Union, 2011\)](#).
14. Nagar, Jay Prakash, Stefano Breschi, and Andrea Fosfuri. "ERC science and invention: Does ERC break free from the EU Paradox?." *Research Policy* 53.8 (2024): 105038.
15. [European innovation scoreboard - European Commission \(europa.eu\)](#)
16. Hyeon BH, Kim HY, Lee CM, Moon SH, Kim ES, Lee SY. Bio-Vision 2016: the second national framework plan for biotechnology promotion in Korea. *Biotechnol J*. 2008 May;3(5):591-600. doi: 10.1002/biot.200700248. PMID: 18446865.
17. [ICT ministry holds meeting on bio venture firms | Yonhap News Agency \(yna.co.kr\)](#)
18. [South Korea: Act to Promote Green Bio Industry Published | Library of Congress \(loc.gov\)](#)
19. [Press Releases - 과학기술정보통신부 > \(msit.go.kr\)](#)
20. [Israel - Public Spending On Education, Total \(% Of GDP\) - 2024 Data 2025 Forecast 1973-2020 Historical \(tradingeconomics.com\)](#)
21. [Ministry of Innovation determines Israel’s national R&D priorities Ministry of Foreign Affairs \(www.gov.il\)](#)
22. [What’s the secret to Switzerland’s status as a global talent hub? | World Economic Forum \(weforum.org\)](#)
23. Rodier, C., et al. "New drug approvals in six major authorities 2010–2019: focus on facilitated regulatory pathways and internationalisation." *R&D Briefing* 77 (2020): 1-24.

## List of figures

|                                                                                                                                                                                                                                                                                                                                                                                 |    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>Figure 1:</b> Patent growth compared to economic efficiency per country over the 2015-2020 period for the sector of IB as a whole. The complete table used to produce this figure is visible in Annex 3. ....                                                                                                                                                                | 10 |
| <b>Figure 2:</b> Countries rank in terms of patent intensity (total counts or per GDP) over the observed period and per technological area. Details of the specific patent affiliates (private companies, research centers, universities...) are available per technological field on the dashboard provided with this publication. ....                                        | 11 |
| <b>Figure 3:</b> Revealed technological advantage for the EU and global competitors (AU: Australia; CA: Canada ; CH: Switzerland ; CN: China ; EU: European Union (27 members) ; UK: United Kingdom ; IL: Israel ; JP: Japan ; KR: South Korea ; SG: Singapore ; TW: Taiwan ; US: United States of America).....                                                                | 13 |
| <b>Figure 4:</b> Revealed technological advantage for EU member states. Only countries with at least 10 patents in each industrial biotechnology sub-technology are presented in this figure (AT: Austria ; BE: Belgium ; DE: Germany ; DK: Denmark ; ES: Spain ; FI: Finland ; FR: France ; IE: Ireland ; IT: Italy; NL: The Netherlands ;; PT: Portugal ; SE: Sweden ;). .... | 14 |
| <b>Figure 5:</b> Distribution of patent affiliations by type (company, public sector) for the 25 most patent-productive countries and the EU, presented as a percentage over the observed period. ....                                                                                                                                                                          | 15 |
| <b>Figure 6:</b> private-public patents co-affiliations. ....                                                                                                                                                                                                                                                                                                                   | 16 |
| <b>Figure 7:</b> Representation of international co-affiliations with EU27 countries grouped (top) or individually (bottom). These graphs represent the number of times countries are co-owners on a patent with each other's. ....                                                                                                                                             | 18 |

## Annexes

**Annex 1:** List of benchmark companies used to generate a starting list of CPC codes and keywords. The patent families number were retrieved from the TIM tool in May 2024.

| Company name            | Founded | Patents families | Incorporation name              | Rationale                                                                     |
|-------------------------|---------|------------------|---------------------------------|-------------------------------------------------------------------------------|
| <b>Codexis</b>          | 2002    | 226              | Codexis Inc                     | Develops enzymes for use in pharmaceuticals and other industries.             |
| <b>Oxford Nanopore</b>  | 2005    | 196              | Oxford Nanopore Tech Ltd        | Provides DNA and RNA sequencing technology.                                   |
| <b>Ginkgo Bioworks</b>  | 2009    | 84               | Ginkgo Bioworks Inc             | Designs custom microorganisms for various industries.                         |
| <b>Arzeda</b>           | 2009    | 8                | Arzeda Corp                     | Creates enzymes and proteins through computational design.                    |
| <b>Sphere Fluidics</b>  | 2010    | 19               | Sphere Fluidics Ltd             | Specializes in single-cell analysis and isolation technology.                 |
| <b>Berkeley Lights</b>  | 2011    | 99               | Berkeley Lights Inc             | Develops platforms for cell-based research and therapeutic discovery.         |
| <b>Synthace</b>         | 2011    | 5                | Synthace Ltd                    | Offers a software platform for automating biological experiments.             |
| <b>Biosyntia</b>        | 2012    | 7                | Biosyntia Aps                   | Produces bio-based ingredients using microbial fermentation.                  |
| <b>Benchling</b>        | 2012    | 2                | Benchling Inc                   | Provides cloud-based software for life sciences research and data management. |
| <b>Zymergen</b>         | 2013    | 122              | Zymergen Inc                    | Engineers microbes to produce bio-based materials and chemicals.              |
| <b>Twist Bioscience</b> | 2013    | 76               | Twist Bioscience Corp           | Manufactures synthetic DNA for research and industrial use.                   |
| <b>Telesis Bio</b>      | 2013    | 21               | Telesis Bio Inc ; Codex DNA Inc | Develops DNA and RNA synthesis technologies.                                  |

|                    |      |      |                                                 |                                                                         |
|--------------------|------|------|-------------------------------------------------|-------------------------------------------------------------------------|
| <b>Isoplexis</b>   | 2013 | 20   | Isoplexis Corp                                  | Offers single-cell proteomics solutions for immune profiling.           |
| <b>Zymtronix</b>   | 2013 | 12   | Zymtronix Llc ; Zymtronix Catalytic Systems Inc | Creates enzyme immobilization technologies for industrial biocatalysis. |
| <b>DNA Script</b>  | 2014 | 45   | Dna Script                                      | Develops enzymatic DNA synthesis technology.                            |
| <b>EnginZyme</b>   | 2014 | 4    | Enginzyme Ab                                    | Uses enzymes for sustainable chemical production.                       |
| <b>Inscripta</b>   | 2015 | 93   | Inscripta Inc                                   | Provides genome-editing technology and tools.                           |
| <b>Evonetix</b>    | 2015 | 13   | Evonetix Ltd                                    | Develops silicon-based DNA synthesis platforms.                         |
| <b>Protera</b>     | 2018 | 4    | Protera S R L                                   | Designs proteins for food and industrial applications using AI.         |
| <b>Illumina</b>    | 1998 | 1172 | Illumina Inc                                    | Offers DNA sequencing and genomics technologies.                        |
| <b>Biomillenia</b> | 2014 | 13   | Biomillenia Sas                                 | Develops microbiome-based solutions for healthcare and agriculture.     |

**Annex 1:** List of the keywords and CPC terms and variations in SQL format used to retrieve the patents in each of the technological domains.

| Synthetic/Engineered biology      | Biological data analysis and computational tools | Robotics, high-throughput assays and lab automation:                         | Bioprocess engineering:            | Molecular diagnostics and analytical techniques: |
|-----------------------------------|--------------------------------------------------|------------------------------------------------------------------------------|------------------------------------|--------------------------------------------------|
| Keywords                          |                                                  |                                                                              |                                    |                                                  |
| Mutagenes* OR                     | Sequence analysis~2 OR                           | (automated AND (microscop* OR assay OR *process OR microscop* OR sampl*)) OR | Biosensor* OR                      | Liquid chromato* OR                              |
| Genome engineer*~2 OR             | Sequence library~2 OR                            | Biochip* OR                                                                  | Cellular process*~2 OR             | Bioassay* OR                                     |
| Base edit* OR                     | Data mining OR                                   | Cell sorting~2 OR                                                            | Process analytical technology~2 OR | Gas chromato* OR                                 |
| Biobrick* OR                      | Sequence data OR                                 | Flow channel OR                                                              | Bioreactor* OR                     | ("flow cell*" AND ("sequenc*" OR "DNA")) OR      |
| Biosynthetic pathway*~2 OR        | Gen* data OR                                     | Flow cytometry OR                                                            | Bioprocess* OR                     | Metabolom* OR                                    |
| ("Cell free" AND "*synthesis") OR | DNA sequence OR                                  | (High-content AND (microscop* OR screen)) OR                                 | Biocatalys* OR                     | sequencing OR                                    |
| Cell free system~2 OR             | Artificial intelligence OR                       | (High-throughput AND (microscop* OR phenotyp* OR screen* OR *assay)) OR      | Protein purification~2 OR          | Nanopore OR                                      |
| Cell free assay~2 OR              | Genome assembly OR                               | Lab-on-a-chip* OR                                                            | Scale-up process*~2 OR             | metabonom* OR                                    |
| Chassis organism* OR              | Machine learning OR                              | Liquid handling~2 OR                                                         | Bionanocatalys* OR                 | Spectroscop* OR                                  |
| Genetic engineer* OR              | Deep learning OR                                 | Microfluidic* OR                                                             | Microbe cultivat*~2 OR             | RNA-seq OR                                       |
| genetic modif* OR                 | High-throughput genotyp*~2 OR                    | Microscale OR                                                                | Microbial cultivat*~2 OR           | proteom* OR                                      |
| Genome edit* OR                   | Base calling OR                                  | Multiplex assay~2 OR                                                         | Downstream process*~2 OR           | Genomic variant~2 OR                             |
| Metabolic engineer* OR            | Gene expression profil* OR                       | Nanowell* OR                                                                 | Continuous cult*~2                 | transcriptomic* OR                               |
| Metabolism engineer* OR           | Unique molecular indexing OR                     | Phenotypic screen~2 OR                                                       |                                    | genom* OR                                        |
| CRISPR OR                         | Bioinformatic* OR                                | Phenotype screen~2 OR                                                        |                                    | lipidom* OR                                      |

|                             |                                 |           |  |                            |
|-----------------------------|---------------------------------|-----------|--|----------------------------|
| deep mutational scanning OR | Biological data*~2 OR           | pipet* OR |  | Mass spec* OR              |
| Directed evolution~2 OR     | Data analysis OR                | robot* OR |  | HPLC OR                    |
| DNA assembly OR             | Genomic coordinate OR           | workflow* |  | Fluorescence microscop* OR |
| DNA synthe* OR              | Genome coordinate OR            |           |  | MS spec* OR                |
| Enzyme optimization~2 OR    | Structure protein*~2 OR         |           |  | Biomark* OR                |
| Epigenetic engineer*~2 OR   | proteine structur*~2 OR         |           |  | Structured illumination OR |
| Gene circuit* OR            | Computational bio* OR           |           |  | UPLC                       |
| Gene synthesis OR           | *omics OR                       |           |  |                            |
| RNA engineer* OR            | Genetic sequence OR             |           |  |                            |
| protein engineer* OR        | Gene sequence OR                |           |  |                            |
| Synthetic biology~2 OR      | Genome scale model*~2 OR        |           |  |                            |
| Synthetic genomic* OR       | infer* OR                       |           |  |                            |
| Synthetic DNA OR            | Metabolic model*~2 OR           |           |  |                            |
| Strain engineer* OR         | Metabolism model*~2 OR          |           |  |                            |
| Pathway engineer* OR        | predict* OR                     |           |  |                            |
| Prime edit* OR              | Protein fold*~2 OR              |           |  |                            |
| Synthetic RNA OR            | Sequence variant OR             |           |  |                            |
| Systems biology OR          | Unique molecular identifiers OR |           |  |                            |
| TALENs OR                   | Variant calling                 |           |  |                            |
| Xenobiology OR              |                                 |           |  |                            |
| Xeno nucleic~2 OR           |                                 |           |  |                            |
| ZFNs OR                     |                                 |           |  |                            |
| biosynthetic OR             |                                 |           |  |                            |

|                                     |                |       |       |       |
|-------------------------------------|----------------|-------|-------|-------|
| engineered<br>microorganism~2<br>OR |                |       |       |       |
| bioengineer* OR                     |                |       |       |       |
| biocircuit OR                       |                |       |       |       |
| engineered cell~2<br>OR             |                |       |       |       |
| modified cell~2<br>OR               |                |       |       |       |
| enhanced cell~2<br>OR               |                |       |       |       |
| metabolic cell~2                    |                |       |       |       |
| <b>CPC Terms</b>                    |                |       |       |       |
| C12N*                               | G06N AND G16B* | B01L* | C12M* | C12Q* |
| C40B*                               | G16B*          | C12M* | B01J* | G01N* |
|                                     |                | B01J* | C12P* | G02B* |

**Annex 3:** Compound annual growth rate (CAGR) (%) of patents produced, total patents and total patents normalized on PPP-GDP over the observed period 2015-2020.

| Countries      | CAGR (%) | Total patents | Total patents per billion USD GDP (PPP) |
|----------------|----------|---------------|-----------------------------------------|
| Switzerland    | 0.9      | 310           | 0.53                                    |
| USA            | -0.6     | 5184          | 0.26                                    |
| Denmark        | 5.4      | 76            | 0.24                                    |
| Belgium        | 11.1     | 116           | 0.20                                    |
| South Korea    | 17.6     | 410           | 0.19                                    |
| Israel         | 30.3     | 66            | 0.19                                    |
| Sweden         | -1.7     | 101           | 0.19                                    |
| Netherlands    | -4.5     | 162           | 0.17                                    |
| United Kingdom | 11.8     | 455           | 0.15                                    |
| Singapore      | 27.2     | 77            | 0.14                                    |
| Germany        | 9.9      | 511           | 0.12                                    |
| Canada         | 0.0      | 202           | 0.11                                    |
| Ireland        | 3.1      | 45            | 0.11                                    |
| Japan          | -5.8     | 580           | 0.11                                    |
| Taiwan         | -2.3     | 103           | 0.09                                    |
| EU             | 4.8      | 1459          | 0.08                                    |
| France         | -0.6     | 235           | 0.08                                    |
| China          | 30.1     | 1469          | 0.07                                    |
| Australia      | 17.6     | 45            | 0.04                                    |
| Italy          | 22.9     | 72            | 0.03                                    |
| Spain          | 2.4      | 40            | 0.02                                    |

## **Getting in touch with the EU**

### **In person**

All over the European Union there are hundreds of Europe Direct centres. You can find the address of the centre nearest you online ([european-union.europa.eu/contact-eu/meet-us\\_en](https://european-union.europa.eu/contact-eu/meet-us_en)).

### **On the phone or in writing**

Europe Direct is a service that answers your questions about the European Union. You can contact this service:

- by freephone: 00 800 6 7 8 9 10 11 (certain operators may charge for these calls),
- at the following standard number: +32 22999696,
- via the following form: [european-union.europa.eu/contact-eu/write-us\\_en](https://european-union.europa.eu/contact-eu/write-us_en).

## **Finding information about the EU**

### **Online**

Information about the European Union in all the official languages of the EU is available on the Europa website ([european-union.europa.eu](https://european-union.europa.eu)).

### **EU publications**

You can view or order EU publications at [op.europa.eu/en/publications](https://op.europa.eu/en/publications). Multiple copies of free publications can be obtained by contacting Europe Direct or your local documentation centre ([european-union.europa.eu/contact-eu/meet-us\\_en](https://european-union.europa.eu/contact-eu/meet-us_en)).

### **EU law and related documents**

For access to legal information from the EU, including all EU law since 1951 in all the official language versions, go to EUR-Lex ([eur-lex.europa.eu](https://eur-lex.europa.eu)).

### **EU open data**

The portal [data.europa.eu](https://data.europa.eu) provides access to open datasets from the EU institutions, bodies and agencies. These can be downloaded and reused for free, for both commercial and non-commercial purposes. The portal also provides access to a wealth of datasets from European countries.

# Science for policy

The Joint Research Centre (JRC) provides independent, evidence-based knowledge and science, supporting EU policies to positively impact society



**EU Science Hub**

[Joint-research-centre.ec.europa.eu](https://joint-research-centre.ec.europa.eu)



Publications Office  
of the European Union